Case 2: Deal of the Century

INSTRUCTIONS
Read the case below. Use the spreadsheet "**Case 2 – Deal of the Century.xlsx**," which contains calculations and charts referenced in the questions, as a template for your final answers. Your team will need to perform other calculations, such as duration, and answer the questions at the end of the case. Type the names of all team members at the top of the first worksheet and set the print area to one page wide. Use other worksheets in the same Excel file to show the work underlying your final answers.

In August of 2020, your portfolio management team at Prudent Investment Management (PIM) has been asked to evaluate whether "century" bonds that have maturities of 100 years should be included in PIM's portfolio. PIM, like many pension funds and insurance companies, is considering purchasing century bonds because it must make extremely long-term payments to pensioners and insurance policy holders. Buying century bonds would extend the average maturity and *duration* of its portfolio. If the firm can match the duration of its assets to that of its liabilities, which is currently 40 years, it would be able to meet its future obligations regardless of what happens to interest rates.

As long-term interest rates on 30-year US Treasuries gradually fell from all-time highs over 15% in 1981 to record lows near 1% in 2020, the duration of pension fund and insurance company liabilities increased dramatically. These firms sought new ways to increase the duration of their investments to keep pace with the duration of their liabilities. In an August 2019 issuance of $300M in century bonds by the University of Pennsylvania (Penn), the school newsletter boasted that the bond yield of 3.61% at issuance was "the lowest interest rate recorded for a domestic century bond." It asserted that "Penn's finance team seized this opportune moment in the markets to take advantage of historically low interest rates and overwhelming demand from investors." [1] Much of the demand for century bonds comes from pension funds.

The issuers of century bonds, including Penn, often use the proceeds to finance long-term investment projects, such as buildings and other durable structures. From the issuer perspective, two benefits of century bonds relative to short-term bonds are: 1) knowing financing costs years in advance, rather than paying uncertain future interest rates; and 2) knowing that funds will be available for the duration of investment projects, rather than facing the whims of future bondholders who may decide to withhold funding.

Tables 1 to 3 below shows that universities, firms, and countries issued billions of dollars of century bonds from 1993 through 2020. Interestingly, century bonds have experienced three periods of great popularity, each following closely on the heels of significant global declines in

[1] See *Penn Today,* August 6, 2019, "Record-setting century bond" by Amanda Mott.

interest rates. In the years 1996 and 1997, many corporations issued century bonds, and the first sovereign countries and universities began issuing these bonds. In 2010 to 2012, century bonds became popular among universities. For example, in October 2011 Ohio State University issued $500M in century bonds at a yield of 4.85%, which was 170 basis points (bps) higher than the yield of 30-year US Treasuries (UST) as shown in the "spread" column. Since 2015, many sovereign states have issued large amounts of century bonds, including Austria's June 2020 issuance of bonds with a par of 2.0B Euros and a record low yield of just 0.88% and a small spread of only 88 bps over 30-year German bunds.

Table 1: Selected US Universities Issuing 100-year ("Century") Bonds from 1993 to 2020

Date Issued	Borrower	Credit Ratings	Par in $M	Fixed Coupon	Yield (YTM)	30-yr UST YTM	Spread over UST
Apr-96	Yale Univ.	Aaa/AAA	125	7.375%	7.375%	6.675%	0.700%
Nov-96	MIT	Aaa/AAA	75	7.250%	7.250%	6.650%	0.600%
Jun-97	Boston Univ.	A1/A	100	7.625%	7.625%	6.675%	0.950%
May-11	MIT	Aaa/AAA	750	5.600%	5.600%	4.300%	1.300%
Aug-11	Univ. So. Cal.	Aa1/AA	300	5.250%	5.250%	3.510%	1.740%
Oct-11	Ohio State Univ.	Aa1/AA	500	4.850%	4.850%	3.150%	1.700%
Nov-11	Cal. Tech.	Aa1/AA+	350	4.700%	4.700%	2.900%	1.800%
Feb-12	Univ. Cal.	Aa1/AA	860	4.860%	4.860%	3.210%	1.650%
Mar-12	Tufts Univ.	Aa2/AA-	250	5.017%	5.017%	3.367%	1.650%
Mar-12	Univ. Penn.	Aa2/AA+	300	4.674%	4.674%	3.274%	1.400%
Jun-12	Bowdoin Col.	Aa2/NR	129	4.693%	4.693%	2.693%	2.000%
Apr-13	Hamilton Col.	Aa2/NR	103	4.750%	4.750%	2.770%	1.980%
Apr-14	MIT	Aaa/AAA	550	4.678%	4.678%	3.598%	1.080%
Nov-14	Ohio Univ.	Aa3/A+	250	5.590%	5.590%	3.090%	2.500%
Apr-15	Univ. Cal.	Aa2/AA	500	4.770%	4.770%	2.470%	2.300%
May-16	Wesleyan Univ.	Aa3/AA	250	4.781%	4.781%	2.650%	2.131%
Jul-16	MIT	Aaa/AAA	500	3.885%	3.885%	2.275%	1.610%
Sep-16	Cal. Tech.	Aa2/AA-	150	4.283%	4.283%	2.230%	2.053%
Sep-17	Univ. Virginia	Aaa/AAA	300	4.179%	4.179%	2.804%	1.375%
Aug-19	**Univ. Penn.**	**Aa1/AA+**	**300**	**3.610%**	**3.610%**	**2.460%**	**1.150%**
Sep-19	Rutgers Univ.	Aa3/A+	330	3.915%	3.920%	2.270%	1.650%
Jan-20	Univ. So. Cal.	Aa1/AA	320	3.226%	3.226%	2.040%	1.186%

Notes for Tables 1 to 3: All bond data in these tables come from FINRA, FRED, Investing.com, and news stories. If a bond's initial yield or coupon rate is unavailable, the bond is assumed to be issued at par value. Par amounts, coupon rates, yields, and spreads of sovereign bonds are denominated in the bond's currency (Euro or USD). Estimates of 30-year Treasury yields during the period when the US government did not issue a 30-year bond are based on 20-year Treasury yields plus the term spread between 20-year and 30-year Treasuries at the end of the period (January 2006).

Table 2: Selected Corporations Issuing 100-year ("Century") Bonds from 1993 to 2020

Date Issued	Borrower	Credit Ratings	Par in $M	Fixed Coupon	Yield (YTM)	30-yr UST YTM	Spread over UST
Jul-93	Disney	Aa3/AA-	300	7.550%	7.550%	6.600%	0.950%
Jul-93	Coca-Cola	Aa3/AA-	150	7.455%	7.455%	6.570%	0.885%
Dec-95	20th Century Fox	Baa1/BBB+	150	7.900%	7.900%	6.100%	1.800%
Dec-96	IBM	A1/A+	850	7.125%	7.220%	6.360%	0.860%
Feb-97	JC Penney	A2/A	500	7.375%	7.375%	6.660%	0.715%
May-97	Ford Motor	Baa3/BBB	500	7.700%	7.810%	6.920%	0.890%
Jul-97	FedEx	Baa2/BBB	250	7.600%	7.620%	6.740%	0.880%
Aug-97	Bristol-Myers Sqb.	A2/A+	300	6.875%	6.875%	6.460%	0.415%
Oct-97	Motorola	Baa3/BBB-	300	5.220%	6.910%	6.330%	0.580%
Dec-97	Archer-Dan. Mid.	A2/A	250	6.950%	6.950%	5.970%	0.980%
Feb-98	Citigroup	Baa1/BBB+	250	6.875%	6.940%	5.850%	1.090%
Mar-05	Norfolk Southern	Baa1/BBB+	300	6.000%	6.000%	4.440%	1.560%
Aug-10	Norfolk Southern	Baa1/BBB+	300	6.000%	5.950%	3.650%	2.300%
Jan-14	Cleveland Clinic	Aa2/AA	400	4.858%	4.858%	3.940%	0.918%
Aug-16	NY Presby. Hosp.	Aa3/AA-	250	4.763%	4.763%	2.240%	2.523%
Jul-18	Norfolk Southern	Baa1/BBB+	300	5.100%	5.110%	3.110%	2.000%

Table 3: Selected 100-year ("Century") Sovereign Bonds from 1993 to 2020

Date Issued	Borrower / Currency	Credit Ratings	Par in M	Coupon	YTM	30-yr Bench YTM	Spread over Bench
Jan-96	China / USD	A3/BBB	100	9.000%	9.120%	6.130%	2.990%
Jun-97	Philippines / USD	A3/BBB	100	8.600%	8.740%	6.700%	2.040%
Oct-10	Mexico / USD	Baa1/BBB	1000	6.100%	6.100%	3.750%	2.350%
Apr-15	Mexico / USD	A2/BBB+	1500	4.000%	4.200%	2.550%	1.650%
Aug-15	Belgium / Euro	Aa3/AA	50	2.500%	2.500%	1.321%	1.179%
Mar-16	Ireland / Euro	Baa1/A+	100	2.350%	2.350%	0.847%	1.503%
Apr-16	Belgium / Euro	Aa3/AA	100	2.300%	2.300%	1.027%	1.273%
Jun-17	Argentina / USD	B3/B	2750	7.125%	7.125%	2.770%	4.355%
Sep-17	Austria / Euro	Aa1/AA+	4200	2.100%	2.112%	1.216%	0.896%
Jun-20	Austria / Euro	Aa1/AA+	2000	0.850%	0.880%	0.004%	0.876%

Beyond hedging and liquidity motives, speculation could motivate the parties undertaking century bond transactions. Some investors purchase century bonds with the expectation that long-term interest rates will soon decline, yielding huge profits. Conversely, some bond issuers prefer long-term debt because they believe that long-term interest rates will soon rise, implying that issuing long-term debt at today's low rates will minimize their expected future interest payments. The CFO of Ohio State, Geoffrey Chatas, explained the university's decision to issue century bonds in 2011: "We didn't set out to issue a 'century,' but the opportunity to fix in incredibly low-rate long-term capital was very attractive to us."[2]

Your team at PIM is only considering investing in century bonds with low default risk because of the difficulty in modeling whether extremely distant cash flows will be paid. Thus, you will only buy bonds of issuers who are likely to remain solvent for 100 years. Although most of the recently issued century bonds meet this criterion, some at PIM question whether railroad companies, such as Norfolk Southern, will be relevant in 100 years. In the questions below, you will specifically analyze **the 2019 century bond issued by Penn**. This Penn bond is among the most recently issued publicly traded bonds with extremely low default risk (credit ratings Aa1/AA) that is denominated in US dollars. As of August 2020, the bond has 99 years remaining until it matures in 2119.[3]

Figures 1 and 2 below show the prices and yields of the Penn century bond from the time of its issuance in August 2019 through August 2020. The bond's par value is $100. Note that the yield graph in Figure 2 looks like an inverted version of the price graph in Figure 1. The latter graph shows price movements of as much as 30% in 2020 that coincide with volatility in interest rates and variation in market liquidity.

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[2] See *Wall Street Journal,* October 19, 2011, "Ohio State University Prices a 'Century Bond'" by Katy Burne.
[3] We assume that the Penn bond matures in August 2119—exactly 100 after issuance and 99 years after PIM analyzes the bond—to simplify calculations, even though the bond actually matures in a different month in 2119.

Figure 1: Prices of the Penn century bond from issuance, August 2019, until August 2020

Figure 2: Yields of the Penn century bond from issuance, August 2019, until August 2020

Key Assumptions

In the problems below, assume that the yield curve for Treasury bonds is always flat for all maturities. Although all Treasuries with different maturities yield the same as each other, the level of the yield curve shifts over time, as summarized in the table below. For example, 10-year and 30-year Treasuries both yield 2.46% in August 2019; and both yield 1.19% in August 2020. The yield curve for Penn bonds is also always flat for all maturities, though the level of the curve changes over time, as described below:

Month-Year	Aug-2019	Aug-2020
US Treasury Yields	2.46%	1.19%
Penn Bond Yields	3.61%	2.84%
Penn Spread	1.15%	1.65%

Penn spread is defined as the difference between the level of the Treasury and Penn yield curves.

All interest rates are compounded semiannually. All coupon bonds pay semiannual coupons and have par values of $100. Assume that all bonds are not callable before maturity.

Because of its large endowment and durable academic reputation, it is unlikely that Penn will default. Assume that the Penn bond's cash flows are guaranteed.

Clearly show all calculations in the appropriate worksheets within your Excel spreadsheet.

In Questions 1 and 2, you will analyze how hypothetical interest rate changes in August 2020 affect the value of three bonds in which PIM can invest:

1) *the 3.61% coupon 99-year Penn ("century") bond that matures in August 2119;*
    a. *This bond yields 2.84% in August 2020.*
2) *a 1.19% coupon 30-Year Treasury bond yielding 1.19% in August 2020; and*
3) *a 30-Year 0% coupon (Zero) with a par value of $100, also yielding 1.19%.*

To simplify calculations in Q1 and Q2, assume that the yield spread between Penn bonds and 30-year T-bonds will remain at the August 2020 spread of 1.65%. Thus, if the yield on the 30-year T-bond were to increase to 2.00%, the yield on the Penn bond would increase to 3.65%; and if the 30-year T-bond yield were to fall to 1.00%, the Penn bond yield would fall to 2.65%.

Question 1

Calculate the bond prices and Macaulay durations of the 99-year ("century") Penn bond, the 30-year T-bond, and the 30-year Zero under three possible interest rate scenarios in August 2020:

a. Low: the Penn century bond yields 1.50% (below the August 2020 yield);
b. Medium: the Penn century bond yields 2.84% (the actual August 2020 yield); and
c. High: the Penn century bond yields 4.18% (above the August 2020 yield).

[Hints: The constant spread assumption implies the 30-year Treasury bond will yield –0.15%, 1.19%, and 2.53% in parts (a), (b), and (c), respectively. The flat yield curve assumption implies that the 30-year T-bond and 30-year Zero will always have the same yields.]

Which of the three bonds has a duration that is closest to the duration of PIM's liabilities (40 years)? Does your answer depend on which interest scenario occurs?

Question 2

Only for this question, consider what would happen if all rates increase by 16 bps, e.g., the century bond yield rises from its initial yield of 2.84% to a new yield of 3.00% in August 2020.

a. Use modified duration to predict the percentage price changes of the century bond and the 30-year Zero. Which bond has the larger predicted percentage price change?
b. Compare the predicted percentage price changes with the actual percentage price changes. Which bond's predicted percentage price change is closer to its actual percentage price change?

Question 3

With 20/20 hindsight (pun intended), we know that most century bond issuers from 2010 to 2019, including Penn, did not get the "deal of the century" by selling bonds when they did. What sequence of bonds could Penn have issued to raise money in a cheaper way for the period August 2019 to August 2119? Assume all bonds are issued at par: coupon rate equals yield at issuance.

Question 4

Using yield data from 2019 and 2020, estimate the amount (PV in 2019) that Penn would have saved in interest costs from pursuing your strategy in Question 3 relative to its chosen strategy of issuing a 100-year bond in August 2019. Assume all bonds are issued at par. Does the savings come from the shift in Treasury yields or Penn's changing spread relative to Treasuries?

Question 5

Now consider the perspective of an investor who bought Penn's bond at issuance in August 2019. Compare the one-year return from investing in the Penn century bond in August 2019 and selling it in August 2020 to the return from investing in a one-year bond in August 2019 with the same initial yield as the Penn bond. Which strategy has higher annualized return? How much? Assume semiannual bond coupons are reinvested in the one-year bond at this bond's initial yield.